

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5.	(a)	(i)		Units saved = 3 000 [kWh]		1		1	1	
		(ii)		Substitution: 3 000 (ecf) × 0.2 (1) Saving = [£]600 (1) Award 1 mark for an answer of [£]60 000	1	1		2	2	
		(iii)	I	$\frac{150\,000}{600\text{ ecf}} = 250$ [weeks] (1)		1		1	1	
			II	$\frac{250\text{ ecf}}{52} = 4.8$ [years] (1) Accept 5 [years]		1		1	1	
	(b)	(i)		60 × 0.095 = 5.7 [kWh] Accept 6 [kWh]		1		1	1	
		(ii)		$\frac{3\,000}{5.7\text{ ecf}} = 526.3$ Accept 526 or 527 Accept 500 if 6 [kWh] is used		1		1	1	
	(c)			<u>GWP</u> of CO ₂ is 1 (1) which is lower than methane / methane has a GWP of 25 (1) [so disagree] Alternative: CO ₂ has a lower <u>GWP</u> (1) than methane (1) [so disagree]			2	2		
				Question 5 total	1	6	2	9	7	0